Supplementary Material for "Same Cluster, Different Wallet: Cluster Dependence in Airdrop Sybil Detection, Evidence from the LayerZero Airdrop"

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Section, table and figure numbers marked "of the article" refer to the main article. Code, data and every result are in the repository (`https://github.com/paven86/layerzero_xgboost`; archived at `https://doi.org/10.5281/zenodo.23294882`).

S1 Feature definitions

All 62 model features are listed below by family. Amounts are in ETH unless marked USD; times are Unix seconds (UTC); *blocks* are Ethereum block numbers or spans of blocks. LayerZero features use activity up to 2024-05-01 23:59:59 UTC and Ethereum features transactions before 2024-05-01 00:00 UTC (Section 3 of the article). The extraction queries are in the repository.

S1.1 LayerZero transactions (22)

- `earliest_l0_tx_time`, `latest_l0_tx_time`: Time of the earliest and the latest LayerZero transaction sent by the wallet.
- `l0_tx_time_span`: Latest minus earliest LayerZero transaction time, in seconds.
- `n_l0_source_chains`, `n_l0_source_contracts`, `n_l0_projects`: Numbers of distinct source chains, source contracts and projects among the wallet's LayerZero transactions.
- `n_l0_project_per_source_chain`: `n_l0_projects` / `n_l0_source_chains`.
- `l0_avg_stargate_swap`, `l0_min_stargate_swap`: Mean and minimum Stargate swap value (USD) over the wallet's LayerZero transactions; a transaction without a swap counts as 0.
- `n_eth_interactions`: Number of distinct LayerZero transactions the wallet sent from Ethereum.
- `l0_to_eth_* (12 features)`: The same statistics restricted to LayerZero transactions whose destination is Ethereum: transaction count (`n_l0_to_eth_txs`); time span; numbers of source chains, source contracts, destination contracts and projects; projects per source chain; minimum, mean and maximum Stargate swap value (USD); and mean and maximum native drop (USD value of gas delivered at the destination).

S1.2 Ethereum transactions (9)

- `min_tx_value_out`, `max_tx_value_out`: Minimum and maximum value of the wallet's outgoing transactions.
- `tx_value_per_block_out`: Total outgoing value divided by the span in blocks between the first and the last outgoing transaction (0 if the span is 0).
- `min_tx_value_in`: Minimum value of incoming transactions.
- `num_transactions_in`: Number of incoming transactions.

- `earliest_tx_block_in`: Block of the first incoming transaction.
- `time_span_in`: Span in blocks between the first and the last incoming transaction.
- `indegree_per_block_in`: Number of distinct senders divided by `time_span_in` (0 if the span is 0).
- `cex_in_count`: Number of distinct centralized-exchange addresses that sent ETH to the wallet.

S1.3 Gas provider (11)

Let p be the wallet's gas provider, the address of its first incoming ETH transfer, and a the amount of that transfer.

- `provider_is_labeled`, `provider_is_interactor`, `provider_is_null`: Indicators that p is a labeled service, a LayerZero interactor, or absent.
- `provider_fan_out`: Number of distinct interactors funded by p.
- `provider_total_gas_provision_amount`, and the `max`, `min` and `avg` versions of the same name: sum, maximum, minimum and mean of the amounts p provided to interactors (mean = total / fan-out).
- `provider_is_star_like_attack`: 1 if p funds more than one interactor and is not a labeled service.
- `is_provider`: 1 if the wallet is itself the gas provider of another address.
- `gas_provision_block_number`: Block of the wallet's first incoming ETH transfer.

S1.4 Gas provision tree (17)

Let T be the tree of the unlabeled provision forest that contains the wallet (Section 3.4 of the article), with node set V (interactors and other addresses), $n = |V|$, root r, depth $d(v)$ of node v below r, $D = \max_v d(v)$, children $C(v)$, leaves $L = \{v : C(v) = \emptyset\}$, internal nodes $I = V \setminus L$, and $g(v)$ the ETH that v received in its funding transfers. Let $P = (g(v) : v \neq r,\ g(v) > 0)$ be the provision amounts and $G = \sum P$. Every wallet of a tree carries that tree's features. A wallet funded directly by a labeled service, or with no recorded funding, is the root of its tree; if it funds no other address, the tree has a single node.

- `tree_size`, `total_gas`, `max_depth`: n, G and D.
- `avg_depth`: $\frac{1}{n}\sum_v d(v)$.
- `branching_factor`: $\frac{1}{n}\sum_v |C(v)| = (n-1)/n$.
- `star_like_ratio`: $|\{v : |C(v)| > 1\}| \,/\, n$.
- `balance_factor`: $D - \min_{v \in L} d(v)$.
- `breadth_factor`: $n/(D+1)$, the mean number of nodes per depth level.
- `breadth_to_depth_ratio`: `breadth_factor`$/D$ (0 if $D = 0$).
- `longest_chain_ratio`: D/n.
- `leaf_to_internal_ratio`: $|L|/|I|$ (0 if $I = \emptyset$).
- `leaf_provision_proportion`: $\sum_{v \in L} g(v) \,/\, G$ (0 if $G = 0$).
- `sparsity`: $|\{v : g(v) > 0\}| \,/\, n$.
- `gas_distribution_entropy`: $-\sum_i q_i \log_2 q_i$ with $q_i = P_i/G$.

Table S2: Address-level (random) against gas-provision-tree (group) split, seed 42, test partition. Δ: group minus random.

	F1			AP			AUROC	
Model	Random	Group	Δ	Random	Group	Δ	Random	Group
XGBoost	0.738	0.716	−0.022	0.797	0.764	−0.033	0.976	0.969
LightGBM	0.741	0.715	−0.026	0.804	0.769	−0.035	0.976	0.970
Logistic regression	0.236	0.240	+0.004	0.162	0.167	+0.005	0.834	0.838

- `gas_distribution_skewness`: Biased sample skewness $m_3/m_2^{3/2}$ of P, where m_k is the k-th central moment (0 if $|P| < 2$ or P is constant).
- `leaf_gas_distribution_entropy`, `leaf_gas_distribution_skewness`: The same two statistics over the leaf amounts $(g(v) : v \in L)$.

The pipeline also computes four tree statistics that are not model inputs, including the Gini coefficient of P, which a pre-specified ablation rule removed.

S1.5 Provision chain (3)

- `depth`: $d(w)$ for the wallet w; 0 for a root.
- `chain_length`: Number of funding hops from the wallet back along first funders until a labeled service is reached or no funder is recorded (cycles and self-funding stop the walk); 1 for every IxL wallet.
- `interactors_in_chain`: Number of LayerZero interactors on that path, the wallet included.

S2 Address-level against tree-level splits

Under the address-level split, 599 of the 5,463 test Sybils share a gas provision tree with a training Sybil. Table S2 compares the two splits on the seed-42 partition with the same hyperparameters. The tree models lose 0.022–0.026 in F1 and about 0.03 in AP, while logistic regression is unaffected. The loss is concentrated where funding trees contain several wallets: LightGBM's F1 falls from 0.847 to 0.453 for IxI and from 0.544 to 0.271 for IxE, against 0.738 and 0.746 for IxL.

S3 The superseded union split

An earlier version of the report-level analysis split the union components (gas provision tree ∪ bounty report) directly into training, validation and test, without paired conditions. It collapsed training diversity (Table S3): 8,260 of the 8,924 training Sybils come from one component, and training contains 146 reports against 271 under the tree split. Test F1 falls to 0.250 ± 0.027, AP to 0.190 ± 0.015 and AUROC to 0.748 ± 0.026 (LightGBM, ten splits), but this mixes report dependence with the loss of training diversity, so we do not use it as a measure of report dependence. The paired design of Section 4.1 of the article replaces it.

S4 Hyperparameter searches

Table S4 lists, for each search, the eight configurations with the highest validation F1, the library-default configuration and the selected configuration (∗). Full grids are in the repository.

Table S3: Superseded union split: composition by partition (rows, Sybils, Sybils in the large components, reports represented) and test performance.

Split	Partition	Rows	Sybils	In large comps	Reports
tree	Train	213,046	8,924	4,478	271
tree	Validation	91,305	3,824	1,968	229
tree	Test	130,435	5,463	2,768	239
union	Train	213,046	8,924	8,260	146
union	Validation	91,305	3,824	954	66
union	Test	130,435	5,463	0	99
Union split, test	*Model*	*F1*	*AP*		*AUROC*
seed 42 (3 seeds)	XGBoost	0.271	0.226		0.814
seed 42 (3 seeds)	LightGBM	0.286	0.217		0.784
10 splits (1 seed)	LightGBM	0.250 ± 0.027	0.190 ± 0.015		0.748 ± 0.026
10 splits, tree split	LightGBM	0.709 ± 0.008	0.761 ± 0.007		0.969 ± 0.002

Selected configurations XGBoost: learning rate 0.05, depth 12, row subsampling 0.5, column subsampling 0.8, minimum child weight 10, early stopping after 50 rounds without improvement (1,098–1,214 rounds across the three seeds). LightGBM: learning rate 0.05, 511 leaves, no row subsampling, column subsampling 0.8, L2 regularization 1 (298–315 rounds). Random Forest: 300 trees, 30% of features per split, no depth limit, at least one sample per leaf.

The validation scores of every configuration are in the repository (`results/02_search_xgboost.csv`, `results/02_search_lightgbm.csv` and `results/13_search_random_forest.csv`).

S5 Provision-network ablation by category

Removing the provision-network features has no significant effect on F1 in any category (Table S5; corrected $p = 0.25$–0.98 for the removal of all three families). The point estimates (all features minus the reduced model) are $+0.003 \pm 0.003$ for IxL, -0.033 ± 0.185 for IxI and -0.053 ± 0.056 for IxE, but their spread across splits is as large as the effects, so we draw no category-specific conclusion.

S6 Label provenance

Bounty-accepted list (labels) `data/20240915_final_sybil_list/fcfs_list.csv` holds 151,784 addresses across all chains accepted through LayerZero's bounty, each with the reporter's reward address, a link to the forum report and a timestamp between 2024-05-18 and 2024-05-30. Of our 434,786 interactors, 18,211 are on it; they come from 311 reports by 145 reporters, and the ten largest reports and the ten most prolific reporters supply 35.6% and 44.6% of them.

Initial list LayerZero published its initial list on 18 May 2024 in the `LayerZero-Labs/sybil-report` repository, which has since been deleted. We use the copy captured by the Wayback Machine on 24 May 2024 and retrieved on 6 October 2026, stored as `review_support/initial_list/initial_list.csv` (34,560,871 bytes; SHA-256 `1a643f70056fe0c0b38d4a13468e968a2de0bd18de6bc25ed7139d9fa691a7cc`). It contains 803,093 distinct addresses: 801,932 20-byte EVM addresses and 1,161 32-byte addresses, consistent with a non-EVM chain that LayerZero supports. Four GitHub mirrors first committed between 18 and 20 May 2024 and a separately distributed copy are byte-identical to it. None of its addresses is on the bounty list, as the bounty rules require; 34,545 of our interactors are on it. The list is used only for the analysis of Section 5.3 of the article and not by

Table S4: Hyperparameter searches (validation only). lr: learning rate; sub: row subsampling; col: column subsampling; mcw: minimum child weight; λ: L2 regularization; mf: features per split; leaf: minimum samples per leaf. $*$: selected configuration.

Rank	Configuration	Val. F1	Val. AP	Rounds
XGBoost (74 configurations)				
1	lr 0.05, depth 12, sub 0.5, col 0.8, mcw 10 *	0.7024	0.7534	1,214
2	lr 0.05, depth 12, sub 0.5, col 0.8, mcw 1	0.7018	0.7538	547
3	lr 0.05, depth 10, sub 0.7, col 1.0, mcw 1	0.7003	0.7510	911
4	lr 0.1, depth 10, sub 0.5, col 0.8, mcw 1	0.7000	0.7552	444
5	lr 0.05, depth 12, sub 0.5, col 1.0, mcw 1	0.7000	0.7528	521
6	lr 0.05, depth 12, sub 0.5, col 0.8, mcw 5	0.6999	0.7540	878
7	lr 0.05, depth 10, sub 0.7, col 0.8, mcw 1	0.6995	0.7545	925
8	lr 0.1, depth 10, sub 0.7, col 1.0, mcw 1	0.6993	0.7502	452
72	lr 0.3, depth 6, sub 1.0, col 1.0, mcw 1 (default)	0.6851	0.7369	646
LightGBM (48 configurations)				
1	leaves 511, lr 0.05, sub 1.0, λ 1.0 *	0.7046	0.7561	315
2	leaves 255, lr 0.05, sub 0.7, λ 1.0	0.7046	0.7570	608
3	leaves 63, lr 0.1, sub 0.7, λ 1.0	0.7044	0.7523	1,178
4	leaves 511, lr 0.05, sub 0.7, λ 1.0	0.7041	0.7579	301
5	leaves 1023, lr 0.05, sub 0.7, λ 1.0	0.7030	0.7593	164
6	leaves 63, lr 0.05, sub 0.7, λ 0.0	0.7029	0.7511	2,089
7	leaves 31, lr 0.05, sub 0.7, λ 1.0	0.7026	0.7552	4,625
8	leaves 255, lr 0.1, sub 0.7, λ 1.0	0.7021	0.7525	297
48	leaves 31, lr 0.1, sub 1.0, λ 0.0 (default)	0.6886	0.7386	2,084
Random Forest (24 configurations)				
1	mf 0.3, leaf 1, depth None *	0.7051	0.7506	–
2	mf 0.5, leaf 1, depth None	0.7024	0.7469	–
3	mf 0.7, leaf 1, depth None	0.6994	0.7411	–
4	mf sqrt, leaf 1, depth None	0.6981	0.7465	–
5	mf 0.5, leaf 5, depth None	0.6980	0.7417	–
6	mf 0.7, leaf 5, depth None	0.6973	0.7381	–
7	mf 0.3, leaf 5, depth None	0.6970	0.7416	–
8	mf 0.5, leaf 20, depth None	0.6803	0.7229	–

the labeling pipeline. The repository documents the provenance of every file (`DATA.md` and `review_support/initial_list/README.md`).

S7 Operating points

The table gives the baseline LightGBM's operating points on the test partition (Section 5.3 of the article).

S8 Mimicry controls

The table gives recall under each attack scope of the mimicry test (Section 5.6 and Figure 6 of the article).

S9 Label robustness

The table gives the effect of LayerZero's initial list on the standard labeling (Section 5.3 of the article).

Table S5: Change in F1 when a provision-network family is removed (all features minus the reduced model; mean ± SD over ten splits), the number of splits in which the full model is higher, and the corrected p-value.

Removed	All	IxL	IxI	IxE
Without all provision-network families	0.000 ± 0.008	+0.003 ± 0.003	-0.033 ± 0.185	-0.053 ± 0.056
full model higher; p	6/10; 0.978	9/10; 0.251	5/10; 0.838	1/10; 0.287
Without Gas provider	+0.002 ± 0.003	+0.003 ± 0.003	-0.017 ± 0.050	-0.009 ± 0.012
full model higher; p	8/10; 0.588	8/10; 0.406	5/10; 0.691	2/10; 0.394
Without Gas provision tree	-0.006 ± 0.006	-0.001 ± 0.002	-0.110 ± 0.136	-0.017 ± 0.054
full model higher; p	2/10; 0.223	2/10; 0.688	3/10; 0.361	2/10; 0.718
Without Provision chain	-0.002 ± 0.004	-0.001 ± 0.003	-0.030 ± 0.098	+0.004 ± 0.025
full model higher; p	3/10; 0.594	3/10; 0.824	5/10; 0.726	6/10; 0.849

Table S7: Operating points of the baseline LightGBM on the test partition. FP: false positives; †: precision if initial-list addresses counted as Sybils.

Threshold	Flagged	TP	FP	FP on list	Precision	Precision†
0.950	2,941	2,568	373	66	0.873	0.896
0.900	3,627	3,057	570	111	0.843	0.873
0.850	4,042	3,309	733	170	0.819	0.861
0.800	4,349	3,489	860	232	0.802	0.856
0.750	4,616	3,612	1,004	307	0.782	0.849
0.700	4,908	3,711	1,197	422	0.756	0.842
0.544 (val.)	5,658	3,975	1,683	705	0.703	0.827
0.500	5,888	4,043	1,845	779	0.687	0.819

S10 Feature-family ablation

The table gives the change in F1 when each feature family is removed (Section 5.6 of the article).

S11 Temporal holdout

The table gives the temporal holdout at the 20%, 30% and 40% cutoffs (Section 5.6 of the article).

Table S8: Recall of test Sybils at the fixed validation threshold when the attacker replaces k features, by attack scope (mean ± SD over five repetitions; for random subsets, over 20 subsets). M1: features drawn independently; M2: copied jointly from one non-Sybil address. Wallet-local (25 features): counts, values and numbers of chains, contracts and projects, computed from the wallet's own transactions and excluding timing features; Ethereum-local (7): the Ethereum-side subset. The groups describe the data source of a feature, not the cost of changing it.

Attack scope	$k = 1$	$k = 3$	$k = 5$	$k = 7$	$k = 10$
No attack					0.728
M1 marginal					
All features (SHAP order)	0.384 ± 0.004	0.056 ± 0.002	0.012 ± 0.001	–	0.001 ± 0.000
Wallet-local only (SHAP order)	0.384 ± 0.004	0.111 ± 0.001	0.021 ± 0.002	–	0.004 ± 0.001
Ethereum-local only (SHAP order)	0.497 ± 0.006	0.209 ± 0.003	0.151 ± 0.004	0.127 ± 0.003	–
Funding-graph only (SHAP order)	0.677 ± 0.003	0.600 ± 0.004	0.522 ± 0.002	–	0.479 ± 0.004
Random k of all 62	0.684 ± 0.094	0.573 ± 0.179	0.418 ± 0.192	–	0.239 ± 0.208
Random k of wallet-local (25)	0.660 ± 0.098	0.569 ± 0.159	0.326 ± 0.156	–	0.163 ± 0.128
M2 profile copy					
All features (SHAP order)	0.384 ± 0.004	0.126 ± 0.003	0.039 ± 0.003	–	0.006 ± 0.001
Wallet-local only (SHAP order)	0.384 ± 0.004	0.127 ± 0.003	0.038 ± 0.000	–	0.012 ± 0.002
Ethereum-local only (SHAP order)	0.497 ± 0.006	0.223 ± 0.004	0.153 ± 0.004	0.133 ± 0.002	–
Funding-graph only (SHAP order)	0.677 ± 0.003	0.594 ± 0.003	0.519 ± 0.002	–	0.481 ± 0.004
Random k of all 62	0.684 ± 0.094	0.578 ± 0.173	0.423 ± 0.185	–	0.253 ± 0.201
Random k of wallet-local (25)	0.660 ± 0.098	0.571 ± 0.159	0.333 ± 0.149	–	0.174 ± 0.124

Table S9: Effect of LayerZero's initial list on the standard labeling (LightGBM, three model seeds, seed-42 tree split, thresholds from validation). Clean negatives: initial-list addresses removed from the negative class in training and validation.

LayerZero initial list among the interactors			
Initial-list addresses			803,093
Overlap with the bounty-accepted list			0
Interactors on the list (all labeled non-Sybil)			34,545
Share of the non-Sybil class			8.3%
Test false positives at the validation threshold (0.544)			
False positives			1,683
of which on the initial list			705 (41.9%)
Precision / if the list counted as Sybil			0.703 / 0.827
Model, test set	*F1*	*AP*	*AUROC*
Baseline, full test	0.715	0.769	0.970
Baseline, without initial-list rows	0.763	0.807	0.973
Clean negatives, full test	0.643	0.453	0.961
Clean negatives, without initial-list rows	0.783	0.833	0.976
Initial list relabeled Sybil (exploratory; prevalence 12.1%)	0.741	0.794	0.921

Table S10: Feature-family ablation (LightGBM, ten splits, mean ± SD). ΔF1: change against all features; "F1 lower": splits in which F1 is lower than with all features.

Configuration	Features	Test F1	ΔF1	Test AP	Test AUROC	F1 lower
All features	62	0.709 ± 0.008	–	0.761 ± 0.007	0.969 ± 0.002	–
Without LayerZero transactions	40	0.646 ± 0.006	-0.063 ± 0.006	0.677 ± 0.009	0.938 ± 0.004	10/10
Without Ethereum transactions	53	0.698 ± 0.007	-0.010 ± 0.004	0.744 ± 0.006	0.965 ± 0.002	10/10
Without Gas provider	51	0.707 ± 0.006	-0.002 ± 0.003	0.757 ± 0.006	0.968 ± 0.001	8/10
Without Gas provision tree	45	0.715 ± 0.005	+0.006 ± 0.006	0.765 ± 0.007	0.969 ± 0.002	2/10
Without Provision chain	59	0.711 ± 0.007	+0.002 ± 0.004	0.761 ± 0.006	0.969 ± 0.002	3/10
Without all provision-network families	31	0.708 ± 0.005	0.000 ± 0.008	0.761 ± 0.006	0.969 ± 0.001	6/10

Table S11: Temporal holdout: models trained on earlier LayerZero cohorts and tested on the latest 20%, 30% and 40% of addresses by first activity. Same-cohort reference: the model trained across cohorts (seed-42 tree split), scored out of sample on the late-cohort rows of its own test partition, a subset of the rows the temporal model is scored on.

				Temporal model				Same-cohort reference			
Late	From	Sybil early	Sybil late	F1	Recall	AP	AUROC	F1	Recall	AP	AUROC
20%	2023-12-17	5.16%	0.29%	0.000	0.000	0.044	0.742	0.494	0.344	0.449	0.953
30%	2023-08-20	5.55%	1.02%	0.166	0.093	0.280	0.888	0.735	0.697	0.784	0.983
40%	2023-06-27	5.11%	2.80%	0.213	0.143	0.256	0.911	0.724	0.776	0.810	0.985